

Combinations



1

a 120

b 17280

c 462

d 3876

e 5

f 1260

g 21

h 1

i 1

j 35

k 10

l $\frac{6}{5}$

2

a $(n + 1)!$

b $n + 1$

c $n(n - 1)$

d $\frac{1}{(n + 5)(n + 6)}$

3

a $n = 7$

b $n = 3, 4$

4

a No solution provided

b No solution provided

c No solution provided

d No solution provided

e No solution provided

f No solution provided

Applications

5

a

	<i>C</i>	<i>P</i>	<i>W</i>
<i>B</i>	<i>C, B</i>	<i>P, B</i>	<i>W, B</i>
<i>G</i>	<i>C, G</i>	<i>P, G</i>	<i>W, G</i>
<i>R</i>	<i>C, R</i>	<i>P, R</i>	<i>W, R</i>
<i>Y</i>	<i>C, Y</i>	<i>P, Y</i>	<i>W, Y</i>

b 12

6 924

7 20 475

8 3 108 105

9 2925



11 126

12 42 504

13 35

14 128

15
a 1820

b 66

16 125 970

17 91

18 6006

19 7350

20 27 720

21
a 4512 b 17 296 c 21 808

22 211 926

23
a 252 b 210 c 126

24 128

25 16 777 216

26
a 15 b 63

27 20

28



29 120

30

a 20

b 6

c 30

31

a 150

b 310

32 30

33 126

34

a 5

b 9

c ${}^nC_2 - n$

35 76

36 672

37 435

38

a 1540

b 120

c $\frac{6}{77}$

39

a 475 020

b 12 376

c $\frac{34}{1305}$

40

a 10 626

b 5

c $\frac{5}{10\,626}$